Pasteurizer



Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 4 labs

Covers these lab items: Bottle flesh pasteurizer; Fruit juice pasteurizer; Pasteurizer; Small Pasteurizer

What it is used for

A pasteurizer heats suitable liquids or food products to a controlled time and temperature combination to reduce harmful microorganisms. Correct process settings, hygienic handling, and prompt cooling where required help maintain food safety and product quality.

Main parts

- Heating vessel or product chamber holds the material being processed.
- Heating element or heat exchanger transfers heat to the product.
- Temperature sensor measures product or process temperature.
- Controller regulates the heating cycle and target temperature.
- Timer controls the required holding period on equipped models.
- Agitator circulates the product in models designed for mixing.
- Inlet and outlet valves control product flow.
- Cooling system or heat exchanger cools the product where provided.

Before you start

1. Read the manual and use a validated process suitable for the specific product.
2. Confirm the pasteurizer is clean, sanitised, and suitable for food contact.
3. Inspect seals, valves, pipes, sensors, controls, and the heating system for damage.
4. Check calibration or verification status for temperature and time controls.
5. Prepare the product and confirm batch size is within the equipment's rated capacity.
6. Ensure clean receiving containers and any required cooling equipment are ready.

How to use

1. Switch on the pasteurizer and select the approved product-specific process settings.
2. Load the measured product without exceeding the permitted capacity.
3. Start heating and monitor the temperature according to the validated procedure.
4. Ensure the product reaches the required temperature throughout the specified holding period.
5. Use the agitator only where the model and process instructions require it.
6. Record the batch details, temperature, holding time, and relevant monitoring results.
7. If the required temperature or holding time is not achieved, follow the approved corrective procedure before releasing the batch.
8. Cool the product promptly using the specified method when the process requires cooling.
9. Transfer the processed product into clean, suitable containers and apply the required labels.
10. Switch off the equipment and clean and sanitise product-contact parts according to the manual.

⚠ Safety

- Use only validated time and temperature settings appropriate to the product and intended process.
- Avoid contact with hot product, steam, heated surfaces, and hot pipes.
- Do not open pressurised vessels or loosen fittings while the system is operating.
- Never bypass temperature controls, alarms, or safety interlocks.
- Use suitable PPE and follow chemical instructions when handling cleaning or sanitising agents.
- Do not release a batch if required process records or safety checks are incomplete.

Care & storage

- Clean and sanitise product-contact surfaces after each batch using approved procedures.
- Inspect seals, valves, filters, and pipes for wear, leaks, or residue.
- Maintain temperature sensors and timers according to the calibration schedule.
- Arrange authorised servicing for heating failures, temperature instability, or persistent leakage.

Common mistakes to avoid

- Using an unvalidated temperature and holding time for the product.
- Starting the timer before the product reaches the required temperature.
- Overfilling the vessel or failing to mix when the process requires it.
- Allowing processed product to cool or remain exposed under unsuitable conditions.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.